

Acorn Instrument API					Ver 1.0				
Black Oak Engineering					6-Dec-24				
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<i>Precision Electronic Instruments</i>									
				Recent changes					
Used in:	Bee, BlueJay, Monarchs, environmental controllers, etc.								
General:	Acorn was developed by Black Oak Engineering as a simple, general purpose, human readable, efficient communications data layer for test instruments & controllers. It must be compact enough to fit into small MCUs. Other protocols such as SCPI do not meet these criteria.								
	Shown here is typical case of Host & Device.								
	General syntax rule is that a cmd triplet requires a control character (=, ?, +, -, *) then a possible argument.								
	'#' suffix appended to response means that command is recognized but not supported by device.								
	The protocol can be network-ized by appending a numerical index to each command triplet.								
	For example, 'aqr2=100ms' addresses cmd aqr to device with index 2. Index 0 is the Host.								
	By default, every command is echo responded with upper case as shown. Echo can be turned off.								
	If there is a problem with a cmd, error text is returned as 'CMD!...'. Error message text should be plain and succinct.								
	In general physical units must always be appended when applicable.					Monarch Mag3			
	Default buffer length is 41 bytes. There is a cmd to ask what is your buffer length.					FW Ver 0.4			
	Single quotes are part of syntax; they denote strings with possible spaces.								
	Baudrate 115200 8N1 is default. Lines terminated with CR/LF. There is a cmd to alter this.					Adaptation: ch1 = x, ch2 = y, ch3 = z			
License:	Released under MIT License.								
	Acorn API is not intended for life support functions.								
	No warranty is made as to its suitability for any particular application.								
Triplet	Syntax	Description, example	Typical response	Note	Support?				
	Data acquisition								
aqm	aqm?3	What is the device-specific measurement on Channel 3?	AQM3=0.56G	[appro unit]. If there is only one channel, index ignored.	yes				
	Device								
dvc	dvc?	Basic AT cmd	DVC='Bee'	Type of device.	yes				
dre	dre+	Reset system.	DRE?		yes				
dmo	dmo?	Model number.	DMO='BE-BLUJ'		yes				
dco	dco?	Contact info	DCO='www.blackoakeng.com'		yes				
dte	dte?	Internal temperature.	DTE=25.0oC		yes				
dvf	dvf?	What is your firmware version?	DVF='1.0.1'		yes				
dvh	dvh?	What is your hardware version?	DVH='1.3a'		yes				
dva	dva?	What is your Acorn API version?	DVA='1.0'		yes				
	Function Abstraction								
x00	xDD'=arg1,arg2...' Sachem-Navi	Execute special cmd DD for instrument.	x00?	May be 0 or variable number args.	X00?			What is the Product ID? Nominal is 0x30.	